

Project Module 3: Robustness Testing & Safe Deployment

Objective:

In this module, you will evaluate how your existing model behaves under small input changes, test its robustness, and propose a simple defense strategy.

You will build on:

- Your **Module 1 dataset and preprocessing notebook**
 - Your **Module 2 model and analysis**
-

Data and Model Reuse Requirement

For this module, you must:

- Use the **same dataset** you selected in Module 1 (German Credit or LSAC)
- Use the **same preprocessing pipeline**
- Use the **same model (or similar model family)** from Module 2

Do not switch datasets or build a completely new pipeline.

What you will do

You will:

1. reuse your baseline model,
 2. evaluate it under input perturbations,
 3. apply one simple defense,
 4. compare performance and fairness before and after,
 5. recommend whether the model is safe for deployment.
-

****** You are encouraged to use Large Language Models (LLMs) to support your thinking and learning. If you do so, you must include a **public link to the conversation(s)** used.
AUC's academic integrity rules apply: submitted work must reflect your own understanding.

Deliverables:

1. Completed notebook
 2. Short technical report
 3. Written answers must reference specific notebook outputs
-

You are expected to reuse your existing code from Modules 1 and 2 and may reference, adapt, or integrate relevant code from Labs 3 and 4 and any provided assignment notebooks where appropriate.

Tasks:

Part 1: Baseline model and clean evaluation (10%)

Reuse your previous pipeline:

- Load data using your Module 1 notebook
- Train your chosen model
- Report:
 - Accuracy
 - Precision / Recall / F1-score

Include in your notebook:

- Model training and evaluation outputs
- Clean test performance

Answer:

1. What is the baseline performance of your model?
 2. Why is performance on clean data not sufficient to trust a model?
-

Part 2: Robustness stress testing (30%)

Create **at least 3 different perturbations** to your dataset.

Examples:

- Add small noise to numeric features

- Round values (reduce precision)
- Introduce missing values and re-impute
- Modify one feature to simulate user manipulation

Evaluate your model on each perturbed dataset.

You must report:

- Performance under each perturbation
- Comparison with baseline performance

Include in your notebook:

- A results table comparing all scenarios
- At least one visualization

Answer:

1. Which perturbation affected performance the most?
 2. Did small changes cause large prediction differences?
 3. What does this suggest about model reliability?
-

Part 3: Fairness and explanation stability (20%)

Using your previous results:

Analyze:

- Whether fairness metrics changed under perturbations
- Whether certain groups are more affected

Check:

- Whether important features changed (Module 2)
- Whether individual predictions became unstable

Include in your notebook:

- Comparison of fairness metrics
- Supporting plots or tables

Answer:

1. Did robustness issues disproportionately affect any group?
 2. Did explanations remain consistent under perturbations?
 3. Why does instability create ethical concerns?
-

Part 4: Defense implementation (15%)

Implement **one simple defense**, such as:

- Clipping extreme values
- Improved imputation
- Removing unstable features
- Training with noisy/augmented data

Evaluate again on:

- Clean data
- Perturbed data

Include:

- Implementation code
- Before/after comparison table

Answer:

1. Did the defense improve robustness?
 2. Did performance decrease?
 3. Why is this trade-off expected?
-

Part 5: Analysis and recommendation (25%)

Provide an evidence-based analysis.

Your discussion must address:

1. Is your model robust enough for real-world use?
2. Which risks remain?
3. Did your defense meaningfully help?
4. Which applications require stronger robustness?
5. Would you deploy this model? Under what conditions?